ABSTRACT

B* Search is an advanced heuristic search algorithm designed for optimal decision-making in adversarial environments, particularly in game-playing scenarios such as chess and strategic planning systems. Unlike traditional search algorithms, B* Search focuses on efficiently exploring the most promising moves while minimizing computational overhead. It operates using interval-based evaluation, where each node is assigned upper and lower bounds that represent the possible outcomes of a move. This allows the algorithm to determine the best move by selectively expanding only those nodes that have the highest potential to influence the final decision.

The primary objective of B* Search in optimal move exploration is to improve decision accuracy while reducing unnecessary search space traversal. By maintaining and updating optimistic and pessimistic estimates, the algorithm ensures that exploration is guided toward moves that are more likely to yield optimal results. This makes it particularly effective in environments with uncertainty and incomplete information. Compared to traditional algorithms such as Minimax and Alpha-Beta pruning, B* Search provides a more focused and efficient exploration strategy.

This study explores the implementation of B* Search for identifying optimal moves in complex problem spaces. The approach involves representing the problem as a decision tree, applying heuristic evaluation functions, and dynamically updating bounds to guide the search process. The performance of the algorithm is evaluated based on metrics such as search efficiency, computational time, and accuracy in identifying optimal solutions.

Despite its advantages, B* Search has certain limitations, including dependence on the quality of heuristic functions and increased complexity in implementation compared to simpler algorithms. Additionally, maintaining interval bounds can require additional memory and computational resources. However, with proper optimization and heuristic design, these challenges can be effectively managed. In conclusion, B* Search offers a powerful and efficient approach for optimal move exploration in decision-making systems. Its ability to balance exploration and efficiency makes it a valuable technique in artificial intelligence, particularly in game theory and strategic problem-solving applications. Future work may focus on integrating machine learning techniques to enhance heuristic evaluation and further improve the performance of the algorithm in dynamic environments

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B*: Navigating Complexity with Interval-Based Search

INTRODUCTION

A rigorous framework for finding the optimal move — even in the most uncertain and complex decision spaces.

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The Challenge: Finding the Optimal Move

The Stakes

- In complex decision-making — games, control systems, adversarial environments — finding the absolute best move is not just desirable, it's essential.

The Limitation

- Traditional algorithms like minimax and A* rely on single-point estimates, which can fall short of guaranteeing true optimality.

The Scale

- Enormous branching factors make exhaustive search impractical — intelligent pruning and evaluation are required.

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Enter B*: A Leap Forward

Developed by Hans Berliner, B* is a best-first graph search algorithm designed to find the least-cost path from an initial node to any goal node.

Best-First
Prioritises the most promising nodes

Graph Search
Explores structured decision spaces

Distinct from A*
Interval-based, not single-point

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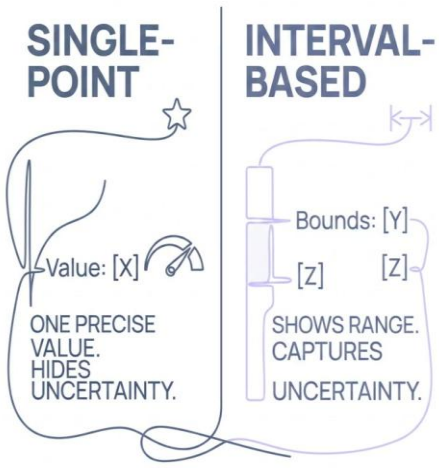
The Core Innovation: Interval Evaluations

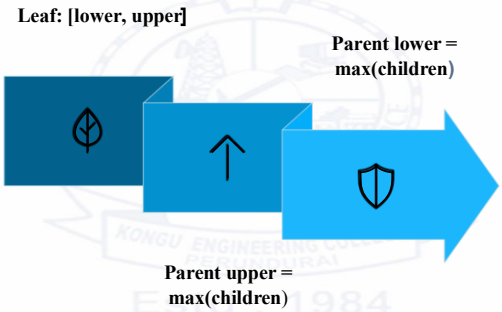
- Rather than assigning a single value to each node, B* assigns an **interval** — a range guaranteed to contain the node's true value.
- This interval-based approach provides a **more robust representation of uncertainty**, enabling the algorithm to reason about what it does and doesn't know.

Each leaf node's interval is guaranteed to contain the true value — a formal bound, not a heuristic guess.



How B* Backs Up Intervals

Interval values propagate upward through the search tree, carrying both potential and uncertainty from leaves to root.



- **This propagation ensures that every parent node reflects the most optimistic and pessimistic possibilities of its subtree — a principled way to manage uncertainty across the tree.**

Achieving "Separation": The Path to Proof

Expand

- **Systematically expand the most promising nodes in the tree**

Compare

- **Check if any child's lower bound meets or exceeds all others' upper bounds**

Prove

- **Separation achieved — the best child is definitively optimal, no further search needed**

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Prove-Best vs. Disprove-Rest

B* employs two complementary strategies to drive the search towards separation efficiently.

Prove-Best

Expands the node with the **highest upper bound**. Goal: raise its lower bound above all competitors, *proving* it is the best move.

Disprove-Rest

Expands the node with the **second-highest upper bound**. Goal: reduce its upper bound below the best child's lower bound, *eliminating* it from contention.

- Strategic selection between these two approaches minimises unnecessary expansion, keeping the search focused and efficient.

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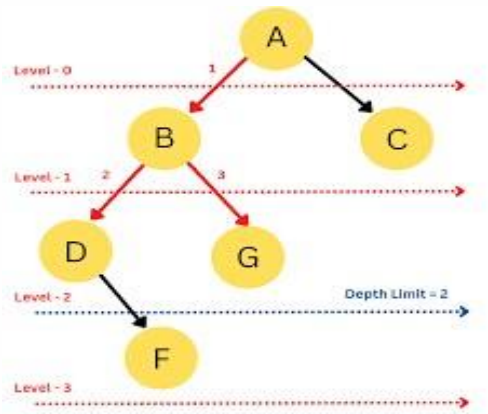

When Limits Are Reached

In highly complex problems, B* may not terminate within practical time or memory constraints. This is an inherent challenge of optimal search.

Artificial Termination

Cutoff criteria are applied to halt search when resources are exhausted

Heuristic Judgment

The best available interval evidence guides the final move selection

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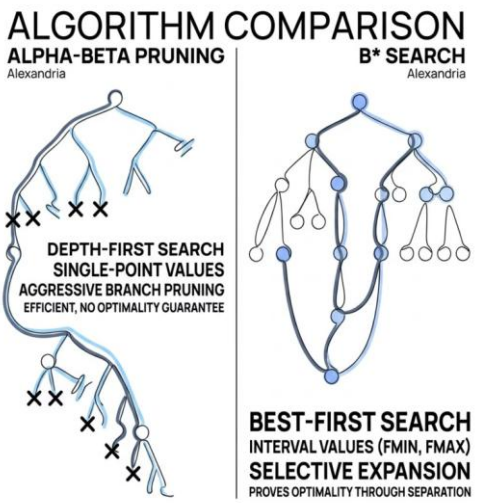
B* vs. Alpha-Beta Pruning



Both algorithms aim to reduce search effort, but their philosophies differ fundamentally.

Alpha-Beta

Depth-first, prunes branches using single-point values. Highly efficient, but cannot formally *prove* optimality.

B*

Best-first, uses intervals to *prove* one move is definitively superior. More rigorous, selectively exploratory.

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The Power of B*: Guaranteed Optimality

Rigorous Guarantee

When separation is achieved, B* **mathematically proves** the selected move is optimal — not just likely.

Manages Uncertainty

Interval evaluations capture what is unknown, enabling **principled reasoning** under incomplete information.

AI Evolution

B* represents a **sophisticated evolution** in search algorithms, pushing the boundaries of what AI can formally guarantee.